
Subject: Remedial Biology (BP106RBT)

Marks: 35

Sem.:- I

Time:- 2 Hr.

Date:-30/12/2017

Timing: 10.00am. – 12:00 pm.

Instructions: - All questions are compulsory.

Q.1 Answer any ONE.

(1 X 10 = 10 Marks)

- A) Define blood. Give composition of blood. Add note on blood group.
- B) Describe parts of typical flowering plant.
- C) Explain different types of connective tissues.

Q. 2. Answer any FIVE

(5 X 05 = 25 Marks)

- A. Describe various functions of lymph node and spleen
- B. Classify tissue.. Explain any 4 types epithelial tissue.
- C. Enlist cranial nerves. Add short note on any 4 cranial nerves.
- D. Discuss functions of digestive system and kidney.
- E. Explain factors affecting plant growth.
- F. Draw a neat well labeled diagram of cell. Describe functions of various cell organelles.
- G. Write a note on structure of heart.

Winter Semester Examination, December – 2017

Subject: Remedial Maths(BP106RMT)

Sem.:- I

Date:-30/12/2017

Marks: 35

Time:- 1:30 Hr.

Timing: 10.00am. – 11:30 pm.

Instructions: - All questions are compulsory.

Q.1 Answer any one.

(1 X 10 = 10 Marks)

- A) solve the differential equation $dy/dx = x-y/(x+y)$
- B) By using the basic result find inverse of laplace transform $4x/x^4+x^2+1$
- C) Using calyhamilton theorem find A^{-1}

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 5 \end{pmatrix}$$

Q. 2. Answer any FIVE

(5 X 05 = 25 Marks)

- A. Find the general solution of differential equation $(2+3\cos x) dy/dx = \sin x$
- B. Find the angle between the line $5=3x+5y$ and $5x+3y=10$
- C. Find equations of sides of triangle A(0,0) B(2,4) C(6,2)
- D. Find laplace transformation $e^{5x}+6x^3-\sin 4x+3\sin 3x$
- E. Show that $x/y + y/x = 7$ if $\log(x+y) = \frac{1}{2} \log x + \frac{1}{2} \log y$
- F. If $a^2=b^3=c^4=d^5$ show that $\log_d abc^3 = 31/5$
- G. If $\log x/(b-c) = \log y/(c-a) = \log z/(a-b)$ show that $x^a y^b z^c = 1$
- H. Solve partial fraction $x^2+5/(x^2+2)(x^2-1)(x^2+2)$
- I. 7. $\lim_{x \rightarrow 0} 2\sin x - \sin 2x/x^3$
 - i. $x \rightarrow 0$
- J. Solve $\int x^4 e^x dx$



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE – RAIGAD

Gourishankar Educational & Charitable Trust's

Gourishankar Institute of Pharmaceutical Education & Research. Limb, Satara

Subject: - Remedial Biology (BP106RBT)

Date: 28/12/2018

Total Marks: 35

Time: 10:00 to 12:30

Q.1. Answer ant ONE

(1X10 = 10 Marks)

- 1) Draw a well labeled diagram of human digestive system & write the function of each organ.
- 2) Explain in detail process of urine formation.

Q. 2. Answer any five

(5X5 = 25 Marks)

- 1) Differentiate between eukaryotic & prokaryotic cells.
- 2) Write short note on nervous tissue.
- 3) Explain three phases of plants growth.
- 4) Discuss structure & function of ovary.
- 5) Discuss anatomy of lungs
- 6) Explain composition of bile.
- 7) Explains process of blood clotting.

Semester I examination

Gouri-shankar college of pharmacy limb

Branch : B Pharmacy

Sem -I

Subject with subject code: Remedial mathematics'

Date: 28/12/2018

Time 10.00-12.30

mark;35

Q.1 solve any one (10)

1. inverse find laplaces transformation of $2s^2 - 4 / (s+1)(s-2)(s-3)$

2. $\int 1/2x^3 + 1 dx$ to 10 parts by Simpsons and trapezoidal rule

Q.2 solve any five

25

1. Solve partial fraction of $x/(x-1)(x+2)(x-3)$

2. Differentiate w.r.t x $e^x + e^x / e^x - e^x$

3. show that $a=b$ if $\log((a+b)/2) = \frac{1}{2}(\log a + \log b)$

4 if $F(x) = x^2 - 3x + 4$ find value of x if $f(x) = F(2x+1)$

5. Find Laplace's transformation $6e^{2t} + 5e^{-3t} + 7t^3 + 2$

6. find A^{-1} if $A = \begin{bmatrix} 2 & 3 \\ 3 & -1 \end{bmatrix}$

7 find the equations of the sides of triangle whose vertices are A (-1,8) B(4,-2) and C(-5,-3)





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Gourishankar Institute of Pharmaceutical Education and Research Limb, Satara

First Semester Theory Examination – 2019-20

Branch: B. Pharm

Sem:-I

Subject: Remedial Biology (BP 106 RBT)

Marks:-35

Date: 28 / 12 / 2019

Time: 10:30 pm to 12:00 pm

Q.1. Answer any ONE

(1 X 10 = 10 Marks)

- 1) Draw a well labeled diagram of human digestive system and write the function of each organ.
- 2) Write a note on male reproductive system.

Q.2. Answer any FIVE

(5 X 5 = 25 Marks)

- 1) Define blood. Give composition of blood.
- 2) Describe any two parts of typical flowering plant.
- 3) Classify tissue. Explain any four types epithelial tissue.
- 4) Write a note on structure of heart.
- 5) Write a structure and functions of lungs.
- 6) Write note about on urinary system and functions of kidney.
- 7) Write note on about on spinal cord physiology.



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Winter Semester Examination 2019

Subject :- Math's

Semester :- I

Marks:- 35

Subject Code :- BP106RMT

Date:- 28/12/2019

Time :- 10.30 to 12.30 pm

Q.1 Solve any One of the following (any 1)

(10 marks)

1. Find A^{-1}

a) $A = \begin{bmatrix} 2 & 8 & 5 \\ 1 & 1 & 1 \\ 1 & 2 & -1 \end{bmatrix}$

b) $B = \begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & -3 \end{bmatrix}$

b) Find Laplace inverse.

$$L^{-1} \left\{ \frac{4S}{S^4 + S^2 + 1} \right\}$$

Q.2 Solve any Five of the following

(5x5=25 marks)

I. Differentiate

a) $y = \log x + \sin^{-1} x$

b) $y = \frac{\sin x}{\log x}$

c) $y = e^x a^x$

II. Evaluate

a) $\lim_{x \rightarrow 0} \frac{\sqrt{1 - \cos 2x}}{x}$

b) $\lim_{x \rightarrow 0} \frac{2x^2 + 9x + 5}{x + 5}$

III. Using Cramer's Rule Solve the following

a) $x + 2y + 3z = 6$
 $2x + 4y + z = 7$
 $3x + 2y + 9z = 14$

IV. Prove that:

1) $\log \frac{33}{18} + \log \frac{4}{7} - \log \frac{27}{21} = 0$

2) $\log \frac{u^2}{vw} + \log \frac{v^2}{wu} - \log \frac{w^2}{vu} = 0$

V. Evaluate

$\int_0^2 \frac{1}{1+x} dx$ correct up to 4 equal part by 1) Trapezoidal 2) Simpson Rule

VI. Find the distance between two points.

a) (2, 3) (4,-1)

b) (-5.7) (-1, 3)

c) (a,b)(-a,b)

VII. Solve by partial fraction

$$\frac{x}{(2x-1)(3x-1)(4x-1)}$$

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RAIGAD-402103

Winter 22 End sem exam

Year: B. Pharmacy (Sem-I)

Subject: -Remedial biology (BP106RBT)

Total Marks: 35

Date: 13/05/2022

Time: 10.00am to 11.30 am

Q. 2. Long answer question (any one)

(1 X 10 = 10 Marks)

1. Explain in detail about female reproductive system. Note on menstrual cycle with neat labeled diagram.
2. Explain human digestive system & role of digestive enzymes..

Q. 2. Short answer question (any five)

(5X 5 = 25 Marks)

1. Write a note on neurone.
2. Define hormone. Give their properties. Classify them with example.
3. Explain the urine formation.
4. Add a note on nutrient uptake
5. Explain function of cerebrum & cerebellum.
6. Explain photosynthesis.

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First Semester Theory Examination WINTER-2022

Semester: I

Class: First Year B. Pharm.

Subject: Remedial Math's (BP106RMT)

Time: 10:00 to 11:30 am

Date:- 13/05/2023

Total marks:- 35

Q. No. 1 Attempt any one

(10 marks)

I) solve partial fraction I) $2x/(x-1)(x-2)(x-3)$ II) $2x/(2x-1)(3x-1)(4x-1)$

II) Verify clayey Hamilton theorem
$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 2 & 2 \end{bmatrix}$$

Q. 2 solve any 5

(25 marks)

1)) solve the equation $X+Y+Z=3$, $2x-y+z=-1$, $2x+y-3z=-9$ by crammer rule

2)) show that the line joining (2,-3) and (-5,1) is parallel to the line joining (7,-1) and (0,3)

3) find Laplace transforms of $6e^{-3t} - 2\cos t + 7t^3 + 2$

4) Partial fraction $x/(x-1)(x-2)^2$

5 show that the point A(1,7) B(4,2) C(-1,-1) and D(-4,-4) are vertices of square.

6) $F(x) = (x-3)/4x-5$, $g(x) = (3+5x)/(4x-1)$ then show that $f \circ g(x) = x$

7) If $a^2 + b^2 = 7ab$ then show that $\log(a+b)/2 = 1/2 \log a + 1/2 \log b$



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First Semester Supplementary Theory Examination Summer-2023

Semester: I

Class: First Year B. Pharm.

Subject: Remedial Math's (BP106RMT)

Time: 1:30 hour

Date:- 05/09/2023

Total marks:- 35

Q. No. 1 Attempt any one

(10 marks)

I) solve partial fraction I) $2x/(x-1)(x-2)(x-3)$ II) $2x/(2x-1)(3x-1)(4x-1)$

II) Verify clayey Hamilton theorem

$$\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 2 & 2 \end{pmatrix}$$

Q. 2 Solve any 5

(25 marks)

1)) solve the equation $X+Y+Z=3$, $2x-y+z=-1$ $2x+y-3z=-9$ by crammer rule

2)) show that the line joining $(2,-3)$ and $(-5,1)$ is parallel to the line joining $(7,-1)$ and $(0,3)$

3) find Laplace transforms of $6e^{-3t} - 2\cos t + 7t^3 + 2$

4) Partial fraction $x/(x-1)(x-2)^2$

5 show that the point $A(1,7)$ $B(4,2)$ $C(-1,-1)$ and $D(-4,-4)$ are vertices of square.

6) $F(x) = (x-3)/4x-5$ $g(x) = (3+5x)/(4x-1)$ then show that $fog(x)=x$

7) If $a^2 + b^2 = 7ab$ then show that $\log(a+b)/2 = 1/2 \log a + 1/2 \log b$



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End Semester Theory Examination WINTER-2023

Semester: I

Class: First Year B. Pharm.

Subject: Remedial Math's (BP106RMT)

Time: 1:30 hour

Date:- 01/ 01/2024

Total marks:- 35

Q. No. 1 Attempt any one

(10 marks)

A) I) show that the point A(1,7) B(4,2) C(-1,-1) and D(-4,-4) are vertices of square.

II) $F(x) = (x-3)/4x-5$ $g(x) = (3+5x)/(4x-1)$ then show that $fog(x) = x$

B) Verify clayey Hamilton theorem

$$\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & 2 & 2 \end{pmatrix}$$

Q. 2 solve any 5

(25 marks)

1) Solve the equation $X+Y+Z=3$, $2x-y+z=-1$ $2x+y-3z=-9$ by crammer rule

2) Show that the line joining (2,-3) and (-5,1) is parallel to the line joining (7,-1) and (0,3)

3) Find Laplace transforms of $6e^{-3t} - 2\cos t + 7t^3 + 2$

4) Partial fraction $x/(x-1)(x-2)^2$

5) Find dy/dx I) $y = e^x + 1/e^x - 1$ II) $y = e^{\sqrt{x}} \sin x$

6) Solve partial fraction I) $2x/(x-1)(x-2)(x-3)$ II) $2x/(2x-1)(3x-1)(4x-1)$

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RAIGAD -402 103**

Gourishankar Institute of Pharmaceutical Education and Research, Limb, Satara.

End Semester Theory Examination – January- 2024

Branch: B.Pharm

Sem.:- I

Subject: Remedial Biology (BP106RBT)

Marks:35

Date: -01/01/2024

Timing: 2:00 Pm. – 3:30 P.m.

Q.1.Attempt any one of the following

(1×10= 10 Marks)

1. Draw neat and labelled diagram of female reproductive system and write a note on anatomy and physiology of female reproductive system.
2. Discuss composition of blood and write a note on ABO blood group system.

Q. 2. Attempt any four of the following.

(5×5= 25Marks)

1. Enlist cranial nerves. Add a short note on any 4 types of cranial nerves.
2. Explain in short general anatomy of root.
3. Enlist classification of five kingdoms and explain one of them.
4. Explain respiration in plants.
5. Discuss structure and function of heart.
6. Write a note on structure and functions of cell and cell organelles.